Damien MAÎTRE

MSc Engineering — Infrastructure, Structural mechanics & Numerical Methods

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ABOUT ME

I am a last year engineering student at the Ecole nationale des ponts et chaussées and at Politecnico di Milano, with hands-on experience in infrastructure engineering and project management. I'm looking for a end of study intenship, for which I would be eager to participate in infrastructure project management, design, construction or engineering, or research on structural mechanics. I'm currently based in Milano, Italy, and I'd be happy to start February or March 2027.

EDUCATION

September 2026 – February 2027
Laurea Magistrale di Ingegneria (MSc)
Student exchange at Politecnico di Milano, Italy
International economy, transport operation, high performance computing, Infrastructure finance & Infrastructure management.

September 2023 – June 2027
MSc Engineering — Fluid dynamics, Structural mechanics, Prototyping
École nationale des ponts et chaussées (ENPC), Paris
Structural mechanics and large-structure design, FEM applied to structural analysis, composite materials, metallurgy, complete fluid dynamics module, HPC, concrete.

- **Tools:** OpenFOAM, StarCCM+, Abaqus, OpenSeesPy, gmsh
- **Projects:** Underwater sea-glider simulation and prototype ; closed-wing aircraft design and optimization ; wooden suspension bridge design, verification and construction.

September 2021 – June 2023
Bachelor — *Classes préparatoires*
Lycée Kléber, Strasbourg
Intensive programme in mathematics, physics and engineering sciences.

RESEARCH PROJECTS

Complete CFD developper-first suite
Personal project — OpenFOAM, Gmsh (forked), Python, C++
Open and user-friendly aerodynamical loads studies, tailored for very high $\mathcal{R}e$ axisymmetric problems.

Structural FEM & Eurocode Verification suite
Personal project — OpenSeesPy, Python
End to end homemade structure design, visualization and verification tools, based on the Eurocode litterature.

PROFESSIONAL EXPERIENCE

February 2026 – July 2026
Infrastructure engineering internship — TRACTEBEL FRANCE
CDG Express project and more, Saint-Ouen

- Construction compliance verification for the CDG Express, a 2.5B€ high-speed rail project,
- Structural analysis and Eurocode verifications of civil engineering structures,
- Review of structural calculation reports and as-built site surveys,
- Drafting technical reports and recommendations for several clients

July 2025 – January 2026
Digital Twin PM intern — SNCF Voyageurs
Industry & digital solutions, Saint-Denis

- Management of a 150k€ digital project within the rolling stock maintenance industry,
- Coordination of 15 3D artists across the country's maintenance plants,
- Project ownership and reporting to internal clients (plant directors, Industrial Director, etc.),
- End-to-end procurement and tendering process.

TECHNICAL SKILLS

CFD & FEM OpenFOAM, StarCCM+, Abaqus, OpenSeesPy, Gmsh
Programming 10+ years of experience with Python, C++ and UNIX environments

LANGUAGES

French — Native · **English** — Proficient (C1) · **Italian** — Working proficiency (B2) · **German** — Working proficiency (B2)

ACTIVITIES & COMMUNITY INVOLVEMENT

Rescuer — French Red Cross
Volunteer first-aid responder
First-aid trained responder at public events (marathons, festivals, Paris-Plage) and on-call with Paris Fire Brigade.

Director — Student Musical Ensemble
BridgeWay Comédie Musicale, Champs-Sur-Marne
Directed an ensemble of 75 artists and a creative/technical leadership team of 12 with 4.5k€ budget.

General Secretary — Student Association
Union des Elèves de l'Ecole Nationale des Ponts et Chaussées
Management of a team of 15 for professional and social events with 1,000+ attendees, Administrative management of a registered French non-profit (Loi 1901).